Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science andCommerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – II

Roll No.: T001	Name:Soham Acharekar
Class: TYIT	Batch: 01
Date of Assignment: 31/07/2026	Date/Time of Submission: 31/07/2026

2. Describing the Distribution of a Variable:
a. Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.
CODE:

- **Import Required Libraries**

```
import pandas as pd
```

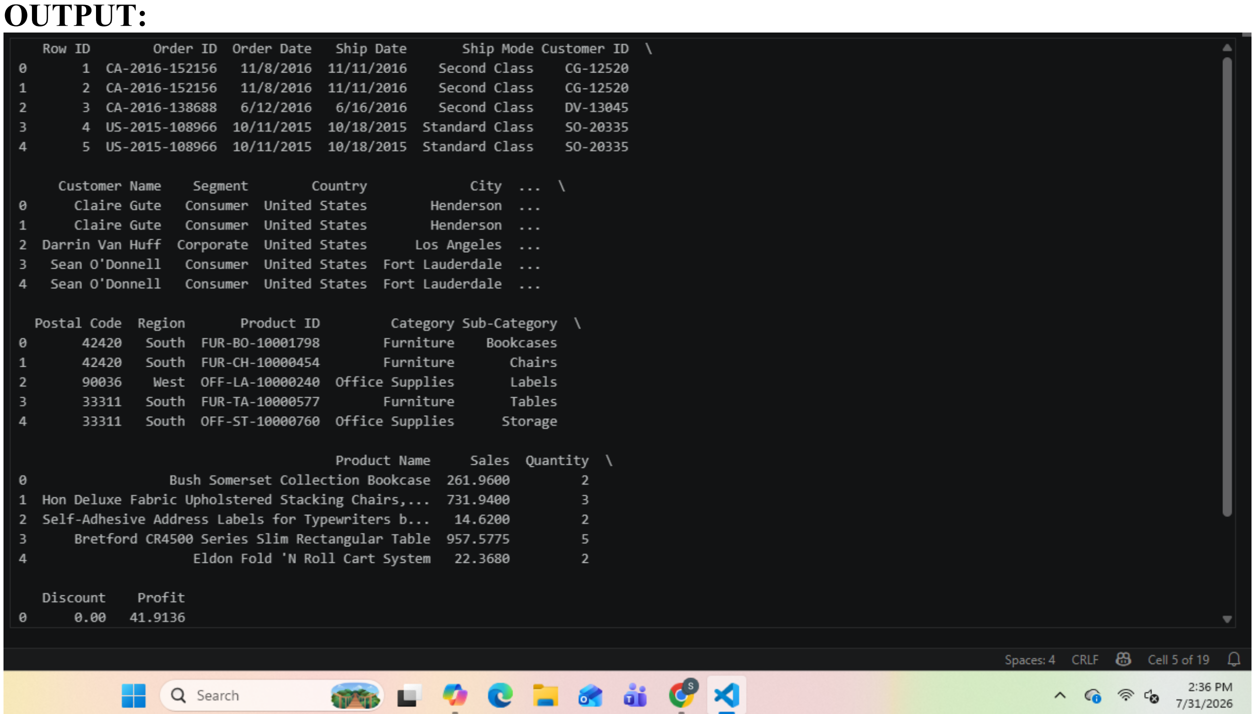
- **Load Dataset**

```
df = pd.read_csv("Sample - Superstore.csv", encoding="latin1")
```

- **Display First 5 rows**

```
print(df.head())
```

OUTPUT:



```
   Row ID        Order ID  Order Date  Ship Date       Ship Mode Customer ID  \
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
1       2  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class    DV-13045
3       4  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases
1       42420   South  FUR-CH-10000454        Furniture       Chairs
2       90036    West  OFF-LA-10000240  Office Supplies       Labels
3       33311   South  FUR-TA-10000577        Furniture       Tables
4       33311   South  OFF-ST-10000760  Office Supplies      Storage

                                        Product Name     Sales  Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600         2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400         3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200         2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775         5
4                     Eldon Fold 'N Roll Cart System   22.3680         2

   Discount    Profit
0      0.00   41.9136
```


- **SALES**

```
sales = df["Profit"]
```

- **DISPLAY MEAN**

```
print("Mean :", sales.mean())
```

OUTPUT:

```
Mean : 28.65689630778467
```

- **DISPLAY MEDIAN**

print("Median :", sales.median())

OUTPUT:

```
Median : 8.6665
```

- **DISPLAY MODE**

print("Mode :")
print(sales.mode())

OUTPUT:

```
Mode :
0    0.0
Name: Profit, dtype: float64
```

- **DISPLAY MINIMUM**

print("Minimum :", sales.min())

OUTPUT:

```
Minimum : -6599.978
```

- **DISPLAY MAXIMUM**

print("Maximum :", sales.max())

OUTPUT:

```
Maximum : 8399.976
```

- **DISPLAY RANGE**

print("Range :", sales.max() - sales.min())

OUTPUT:

```
Range : 14999.954000000002
```

- **DISPLAY VARIANCE**

print("Variance :", sales.var())

OUTPUT:

```
Variance : 54877.79805537892
```

- **DISPLAY Standard Deviation**

print("Standard Deviation :", sales.std())

OUTPUT:

```
Standard Deviation : 234.2601076909573
```

- **DISPLAY COUNT**

print("Count :", sales.count())

OUTPUT:

```
Count : 9994
```

- **DISPLAY SUM**

print("Sum :", sales.sum())

OUTPUT:

```
Sum : 286397.0217
```

- **DESCRIPTIVE STATISTICS**

print("\nDescriptive Statistics")
print(sales.describe())

OUTPUT:

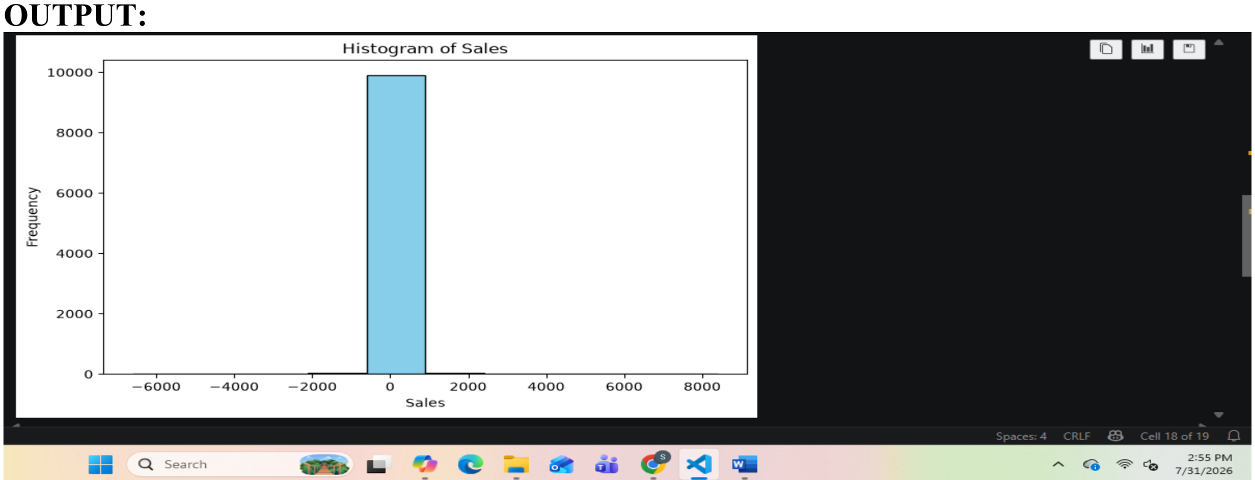
```
Descriptive Statistics
count     9994.000000
mean        28.656896
std        234.260108
min      -6599.978000
25%          1.728750
50%          8.666500
75%         29.364000
max       8399.976000
Name: Profit, dtype: float64
```

b. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

CODE:

```
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8,5))
plt.hist(sales, bins=10, color="skyblue", edgecolor="black")
plt.title("Histogram of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT:



CODE:

```
plt.figure(figsize=(8,3))
sns.boxplot(x=sales, color="orange")
plt.title("Box Plot of Sales")
plt.xlabel("Sales")
plt.show()
```

OUTPUT: